ELEC40012 - MATHEMATICS1A - QUESTIONS

Section A: Multiple Choice

Enter your answers on the multiple-choice answer sheet. Please note that some of the questions are worth 2 marks, and some 3 marks. Total marks for this section: 40.

1. Given complex $z = x + iy$, which of the following are the real and imaginary parts of $w = \cos(iz^2)$?

   (a) $\mathrm{Re}(w) = -\cosh(2xy)\cos(x^2-y^2)$, and $\mathrm{Im}(w) = \sinh(2xy)\sin(x^2-y^2)$,

   (b) $\mathrm{Re}(w) = \cosh(2xy)\sin(x^2-y^2)$, and $\mathrm{Im}(w) = \sinh(2xy)\cos(x^2-y^2)$,

   (c) $\mathrm{Re}(w) = \sin(2xy)\cosh(x^2-y^2)$, and $\mathrm{Im}(w) = -\cos(2xy)\sinh(x^2-y^2)$,

   (d) $\mathrm{Re}(w) = \cos(2xy)\cosh(x^2-y^2)$, and $\mathrm{Im}(w) = \sin(2xy)\sinh(x^2-y^2)$,

   (e) none of the above. [3]

   Solution Begin with

$$\cos(iz^2) = \cos[i(x^2-y^2+2xyi)] = \cos(i(x^2-y^2)-2xy) = \cos[i(x^2-y^2)]\cos(2xy) + \sin[i(x^2-y^2)]\sin(2xy)$$

   Now use relationship between trigonometric and hyperbolic functions to write

$$\cos(iz^2) = \cosh(x^2-y^2)\cos(2xy) + i\sinh(x^2-y^2)\sin(2xy)$$

   giving real and imaginary parts as in (d).

2. If n is any integer, which of the following is the locus $z = x + iy$, in the complex plane, satisfying $Arg(z-i) = \dfrac{\pi}{4} + n\pi$?

   (a) the line $y = x+1$, excluding $(0,1)$, (b) the line $y = -x-1$, excluding $(-1,0)$,

   (c) the line $y = -x+1$, excluding $(1,0)$, (d) the line $y = x-1$, excluding $(0,-1)$,

   (e) none of the above. [3]

   Solution Let $Z = z - i$, then $Arg(Z) = \pi/4$ or $-3\pi/4$, either way, lying on the line y=x. So $z = Z + i$, a vertical translation, giving the line $y = x+1$. We exclude the point $(0,1)$, where $z = i$, as $z - i = 0$, where Arg is not defined, hence (a).

3. Which of the following statements about $\lim_{x\to\infty} x^q\left(\sqrt{x-2}-\sqrt{x+2}\right)$ is true?

(a) If $q=1$, the limit is equal to 0 ; (b) If $q=-\frac{1}{2}$, the limit does not exist ;

(c) If $q=\frac{1}{2}$, the limit is equal to 2 ; (d) the limit cannot be determined ;

(e) none of the above. [3]

Solution: First write $\sqrt{x\pm 2}=\sqrt{x\left(1\pm\frac{2}{x}\right)}$ and rewrite

$$x^q\left(\sqrt{x-2}-\sqrt{x+2}\right) = x^{q+\frac{1}{2}}\left[\left(1-\frac{2}{x}\right)^{1/2}-\left(1+\frac{2}{x}\right)^{1/2}\right]$$
$$= x^{q+\frac{1}{2}}\left[\left(1+\frac{1}{2}\left(-\frac{2}{x}\right)+\frac{\frac{1}{2}(\frac{1}{2}-1)}{2}\left(-\frac{2}{x}\right)^2+\ldots\right)-\left(1+\frac{1}{2}\left(\frac{2}{x}\right)+\frac{\frac{1}{2}(\frac{1}{2}-1)}{2}\left(\frac{2}{x}\right)^2+\ldots\right)\right]$$
$$= x^{q+\frac{1}{2}}\left(-\frac{2}{x}+\frac{K_1}{x^3}+\frac{K_2}{x^5}+\ldots\right) = -2x^{q-\frac{1}{2}} + \text{higher order terms},$$

for constants K_i. Hence, for $q=1/2$, the limit exists and is equal to -2, hence (e).

4. The limit given as $\lim_{x\to\frac{\pi}{2}}\left(\frac{1-\sin x}{\frac{\pi}{2}-x-\cos x}\right)$

(a) exists, but cannot be determined , (b) does not exist ,

(c) is equal to -2 , (d) is equal to 0 ,

(e) none of the above. [3]

Solution Check that it's "$\frac{0}{0}$" and use l'Hopital, twice:

$$\lim_{x\to\frac{\pi}{2}}\frac{1-\sin x}{\frac{\pi}{2}-x-\cos x} = \lim_{x\to\frac{\pi}{2}}\frac{-\cos x}{-1+\sin x} = \lim_{x\to\frac{\pi}{2}}\frac{\sin x}{-\cos x} .$$

The denominator vanishes, but not the numerator, so the limit does not exist, hence (b).

5. The motion of a particle at time $t>0$, is described by x, with velocity $\frac{dx}{dt}=\frac{\ln t}{t}$. At time t_1, the particle achieves maximal velocity, and at time t_2, the curvature of the velocity graph changes. Hence

(a) $t_1=0$ and $t_2=e$, (b) $t_2=e$ and $t_1=e^{2/3}$,

(c) $t_1=e^{3/2}$ and $t_2=e$, (d) $t_2=e$ and $t_1=1$, ;

(e) none of the above. [3]

Solution The velocity is maximal when its derivative is zero:

$$\left(\frac{\ln t}{t}\right)' = \frac{\frac{1}{t}\cdot t-\ln t\cdot 1}{t^2} = 0 \Rightarrow \ln t = 1 \Rightarrow t_1 = e.$$

The curvature of the velocity graph changes when its second derivative is zero, hence

$$\left(\frac{1-\ln t}{t^2}\right)' = \frac{-\frac{1}{t}\cdot t^2-(1-\ln t)2t}{t^4} = 0 \Rightarrow -1-2(1-\ln t)=0 \Rightarrow \ln t = \frac{3}{2} \Rightarrow t_2 = e^{3/2},$$

so (e).

6. If $\cos x - \sin y = e^{x+y}$, which of the following is $\dfrac{dy}{dx}$?

(a) $-\dfrac{e^{x+y}-\sin y}{\cos x - e^{x+y}}$, (b) $-\dfrac{e^{x+y}+\cos x}{\sin y - e^{x+y}}$, (c) $-\dfrac{e^{x+y}+\cos y}{\sin x - e^{x+y}}$, (d) $-\dfrac{e^{x+y}+\sin x}{\cos y + e^{x+y}}$,

(e) none of the above. [2]

Solution: Differentiate implicitly, giving

$$-\sin x-(\cos y)\frac{dy}{dx} = e^{x+y}\left(1+\frac{dy}{dx}\right) \Rightarrow (e^{x+y})\frac{dy}{dx}+(\cos y)\frac{dy}{dx} = -\sin x - e^{x+y},$$

so (d).

7. If $y = \sin v$, where $v = u + \dfrac{1}{u}$ and $u = \ln x$, what is the value of $\dfrac{dy}{dx}$ at $x = e$?

(a) $\dfrac{1}{e}$; (b) -1 ; (c) $\dfrac{2}{e^2}$; (d) $\sec^2(e)$;

(e) none of the above. [2]

Solution Using Chain rule, we have

$$\frac{dy}{dx} = \cos\left(\ln x + \frac{1}{\ln x}\right)\left[\frac{1}{x} - \frac{1}{(\ln x)^2}\frac{1}{x}\right] = \cos\left(\ln x + \frac{1}{\ln x}\right)\frac{1}{x}\left(1-\frac{1}{(\ln x)^2}\right) = 0$$

given that $\ln(e) = 1$, so (e)

8. If $I(t) = \displaystyle\int_0^t x^2 e^{-2x}\,dx$ is a function of t, which of the following statements is **true**?

(a) $I(t)$ is decreasing for $t > 1$, (b) $I(t)$ has a point of inflection at $t = 1$,

(c) $I(t)$ has a local maximum at $t = 1$, (d) $I(t)$ has a point of inflection at $t = 1 + \dfrac{1}{\sqrt{2}}$.

(e) none of the above . [3]

Solution If $I(t)$ is the integral, then the fundamental theorem of calculus gives

$$\frac{dI}{dt} = t^2 e^{-2t}$$

which has no stationary points other than $t = 0$. However,

$$\frac{d^2I}{dt^2} = 2t(1-t)e^{-2t}$$

indicates a point of inflection at $t = 1$, hence (b).

9. The integral $I = \int \frac{1}{1+\sin x}\, dx$ is equal to

(a) $-\frac{2}{1+\tan(x/2)}+\mathrm{C}$, (b) $\tan x+\sec x+\mathrm{C}$, (c) $\frac{x}{1+\sin x}+\mathrm{C}$, (d) $\frac{2}{\cot(x/2)-1}+\mathrm{C}$,

(e) none of the above. [3]

Solution Multiply integrand appropriately

$$I = \int \frac{1}{1+\sin x}\frac{1-\sin x}{1-\sin x}\, dx \;=\; \int \frac{1}{\cos^2 x}\, dx - \int \frac{\sin x}{\cos^2 x}\, dx \;=\; \tan x - \sec x + C$$

or, better, use substitution $t = \tan(x/2), dx = \frac{2\,dt}{1+t^2}, \sin x = \frac{2t}{1+t^2}$, so

$$I = \int \frac{1}{1+\frac{2t}{1+t^2}}\frac{2\,dt}{1+t^2} \;=\; \int \frac{2}{1+2t+t^2}\, dt \;=\; \int \frac{2}{(1+t)^2}\, dt \;=\; -\frac{2}{1+t}+C \;=\; -\frac{2}{1+\tan(x/2)}+c$$

so (a)

10. Two series are given as

$$S_1 = \sum_{n=1}^{\infty} \frac{1}{n^{1+\varepsilon}} \quad \text{and} \quad S_2 = \sum_{n=1}^{\infty} \frac{x^n}{\sqrt{n}}\,.$$

Which of the following statements is true?

(a) S_1 diverges for $\varepsilon \geq 0$, and S_2 has radius of convergence $R = 1/2$;

(b) S_1 converges for $\varepsilon \leq 0$, and S_2 has radius of convergence $R = 1$;

(c) S_1 diverges for $\varepsilon \leq 0$, and S_2 has interval of convergence $[-1,1)$;

(d) S_1 has radius of convergence $R = 1$, and S_2 diverges for $|x| > 1$;

(e) none of the above. [3]

Solution S_1 diverges for $\varepsilon \leq 0$ and converges for $\varepsilon > 0$, by integral test. So (a) and (b) are nonsense, likewise (d), as S_1 is not a power series. S_2 has interval of convergence $(-1,1)$ by Ratio Test, but also converges for $x = -1$, using alternating series test. Hence (c).

11. What is wrong with the following argument about the Maclaurin series for $f(x) = \sqrt{1+x}$?

To first order with remainder term, the series is given by $f(x) = 1 + \frac{1}{2}x + R_1$, *(1)*

where $R_1 \;=\; \frac{f''(c)}{2}x^2 = \frac{1}{8(1+c)^{3/2}}x^2$ *, with c between 0 and x.* *(2)*

If we estimate $\sqrt{0.9} \approx 1 + \frac{1}{2}(-0.1) = 0.95$, *(3)*

the remainder can be used to give a bound for the maximum error:

$$|E| < \frac{1}{800(0.9)^{3/2}}\,. \qquad (4)$$

(a) the coefficients of the series in (1) are wrong,

(b) the coefficient of the remainder in (2) is wrong ,

(c) the errror in (4) is bounded using the wrong value of c,

(d) there is nothing wrong with the argument ,

(e) none of the above. [3]

Solution $f'(x) = \frac{1}{2}(1+x)^{-1/2}$ so $f''(x) = -\frac{1}{4}(1+x)^{-3/2}$. The remainder term will then be

$$R_1 = \frac{f''(c)}{2}x^2 = -\frac{1}{8(1+c)^{3/2}}x^2 \, .$$

so (b).

12. To solve the linear first order ordinary differential equation (ODE)

$$\sec(x)\frac{dy}{dx} - \frac{2y}{\sin(x)} \; = \; 1$$

we obtain the integrating factor $\mu(x)$:

(a) $\mu(x) = \csc^2(x)$, (b) $\mu(x) = \ln|\tan(x/2)|$, (c) $\mu(x) = -\sin^2 x$, (d) $\mu(x) = e^{-\int \frac{2}{\sin x}dx}$,

(e) none of the above . [2]

Solution In standard form we write

$$\frac{dy}{dx} - \frac{2\cos x}{\sin x}y \; = \; \cos x \, ,$$

and the integrating factor will be

$$\mu(x) = e^{\int P(x)\,dx} = e^{-2\int \frac{\cos x}{\sin x}\,dx} \; = e^{-2\ln\sin x} = \frac{1}{\sin^2 x} \, ,$$

hence (a).

13. To solve the linear first order ordinary differential equation (ODE)

$$\sec(x)\frac{dy}{dx} - \frac{2y}{\sin(x)} \; = \; 1 \, ,$$

satisfying the initial condition $y(\pi/2) = 1$ we use the integrating factor $\mu(x)$ found in the previous question to obtain

(a) $y \; = \; \cos x(2\sin x + 1)$; (b) $y \; = \; \sin x(2\sin x - 1)$;

(c) $y \; = \; \frac{1}{\cos x}(1 - 2\sin x)$; (d) $y \; = \; \frac{1}{\sin x}(2 + \cos x)$;

(e) none of the above. [3]

Solution Multiplying the standard form of the ODE by the IF, we obtain

$$\frac{1}{\sin^2 x}\frac{dy}{dx} - \frac{2\cos x}{\sin^3 x}y \;=\; \frac{d}{dx}\left(\frac{1}{\sin^2 x}y\right) \;=\; \frac{\cos x}{\sin^2 x}\,,$$

so integrating we get

$$\frac{1}{\sin^2 x}y \;=\; \int \frac{\cos x}{\sin^2 x}\,dx \;=\; -\frac{1}{\sin x} + C \Rightarrow y = -\sin x + C\sin^2 x\,,$$

The initital condition $y(\pi/2) = 1$ gives $1 = -1 + C(1^2) \Rightarrow C = 2$, hence (b).

14. The substitution $x = e^t$ transforms the Euler equation

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = 0\,,$$

into which of the following linear second-order ODEs?

(a) $\dfrac{d^2y}{dt^2} - 2\dfrac{dy}{dt} + y = 0\,,$ (b) $\dfrac{d^2y}{dt^2} + 3\dfrac{dy}{dt} + y = 0,$

(c) $\dfrac{d^2y}{dt^2} - 3\dfrac{dy}{dt} + y = 0\,,$ (d) $\dfrac{d^2y}{dt^2} + 2\dfrac{dy}{dt} - y = 0\,,$

(e) none of the above. [2]

Solution See sect B, Q3, where this is derived. The answer is easily found to be

$\dfrac{d^2y}{dt^2} + 2\dfrac{dy}{dt} + y = 0$, so (e).

15. After employing the substitution in the previous question, the solution of the Euler equation

$$x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + y = 0\,,$$

is found to be

(a) $y \;=\; \dfrac{1}{x}(1 + \ln|x|),$ (b) $y \;=\; \dfrac{x}{1 + \ln|x|}$

(c) $y \;=\; x(1 + \ln x) + C\;,$ (d) $y \;=\; \dfrac{1}{x}(\ln x - 1) + C\,,$

(e) none of the above. [3]

Solution Solving the correct equation, $\dfrac{d^2y}{dt^2} + 2\dfrac{dy}{dt} + y = 0$, the auxiliary equation is $\lambda^2 + 2\lambda + 1 = 0 \Rightarrow \lambda = -1$, a repeated root, with solution

$$y = (A + Bt)e^{-t}$$

Substituting back $x = e^t \Rightarrow t = \ln|x|$, we get

$$y = (A + B\ln|x|)x^{-1},$$

so (e).

Section B

Enter your full answers in the answer booklet. This section has three questions, each worth 20 marks.

1. (a) Solve the equation $\cosh x - \frac{3}{2}\sinh x = 1$, for real x.

   Hence solve $\cosh z - \frac{3}{2}\sinh z = 1$, for complex z.

   Hence, or otherwise, solve $\cos x - \frac{3}{2} i \sin x = 1$, for real x. [7]

   Solution (i) Write cosh and sinh as exponentials and simplify:

   $$\frac{e^x + e^{-x}}{2} - \frac{3}{2}\left(\frac{e^x - e^{-x}}{2}\right) = 1 \implies 2e^x + 2e^{-x} - 3e^x + 3e^{-x} = 4$$

   now multiply by e^x and get

   $$e^{2x} + 4e^x - 5 = 0 \implies e^x = 1, -5$$

   Reject $e^x = -5$ for real x, leaving only $e^x = 1$, so $x = 0$ is the only real solution.

   Most people got this, except for arithmetic error.

   (ii) For complex z we write the r.h.s as a complex exponential:

   $$e^z = 1 = e^{i(2n\pi)} \implies z = (2n\pi)i \quad \text{for integer } n\ ,$$

   or

   $$e^z = -5 = 5e^{i(2n+1)\pi} \implies z = \ln 5 + (2n+1)\pi i \quad \text{for integer } n\ .$$

   Again, a lot of good answers. A few people forgot about periodicity.

   (iii) Can write in terms of cosh and sinh, but this would make it a longer calculation. Instead, compare real/imaginary parts:

   $$\cos x - \frac{3}{2} i \sin x = 1 + 0i \Rightarrow \cos x = 1 \quad \text{and} \quad \sin x = 0\ .$$

   The second equation suggests $x = n\pi$ for any integer, but only the even integers also satisfy the first equation, hence $x = 2n\pi$ for any integer n.

   This is an easy question, if you spot it. Quite a few people didn't and tried going into hyperbolic function, tying this to the first two parts. Not usually successful, as this made the question much harder.

   (b) Given $f(x) = \dfrac{x^2 - x}{x - 2}$, obtain all stationary points and classify them using the second derivative test. Obtain all asymptotes and intercepts and hence sketch $f(x)$, explaining your reasoning. [7]

Solution: Begin by using polynomial division to write

$$f(x) = x+1+\frac{2}{x-2} \Rightarrow f'(x) = 1-\frac{2}{(x-2)^2} \Rightarrow f''(x) = \frac{4}{(x-2)^3}$$

Set $f'(x) = 0 \Rightarrow (x-2)^2 = 2 \Rightarrow x = 2 \pm \sqrt{2}$ are the stationary points, and the second derivative test gives $f''(2+\sqrt{2}) = 4/\sqrt{2}^3 > 0$, and $f''(2-\sqrt{2}) = 4/(-\sqrt{2})^3 < 0$, so $x = 2+\sqrt{2}$ is a local minimum point, and $2-\sqrt{2}$ a local maximum. It is a good idea to estimate the value of $f(x)$ at the stationary points: $f(2+\sqrt{2}) = 2+\sqrt{2}+1+\sqrt{2} \approx 5.8$ and $f(2-\sqrt{2}) = 2-\sqrt{2}+1-\sqrt{2} \approx 0.2$.

Clearly $f''(x) \neq 0$ for any x, so no points of inflection. Asymptotes are $x = 2$, vertical, and $y = x+1$, diagonal. As $x \to \infty, y = x+1+2/(x-2)$ approaches the diagonal asymptote from above. Likewise, as $x \to -\infty, y = x+1+2/(x-2)$ approaches the diagonal asymptote from below. Near the vertical asymptote, as $x \to 2^+$, from the right, the numerator stays near 1, but the denominator gets smaller, staying positive, so $f(x) \to +\infty$, and likewise, as $x \to 2^-$, from the left, $f(x) \to -\infty$. This is more than enough to sketch the function, see sketch at end.

A large minority of people didn't see the diagonal asymptote, or gave it as $y = x$. Almost noone discussed $x \to s^{\pm}$ or $f(x) \to (x+1)^{\pm}$, and only very few people looked for POI. Some people inferred a POI where none exist. Not estimating y at stationary points led to sketches far off the real thing.

(c) Given $f(x)$ as in Q1b, let $g(x) = f(x)[H(x) - H(x-2)]$, where $H(x)$ is the Heaviside function. Sketch $g(x)$, and hence sketch $g_e(x)$ and $g_o(x)$, the even and odd parts of $g(x)$, taking care to label the axes. [6]

Solution We observe that the effect of the Heasidide terms is to take the part of $f(x)$ on the interval $[0,2]$. We can sketch $g(x)$, then reflect into $[-2,0]$ for $g(-x)$ and add/subtract to obtain even and odd parts. Need to label the $y-$axis so that max is at about 0.2 for $g(x)$, and about 0.1 for the even/odd parts, see sketch at end.

A lot of confusion here, even when (b) was done correctly.

2. (a) Obtain the integrals:

$$\int \cosh^{-1} x \, dx \,, \qquad \int_0^1 3x^2 \tan^{-1} x \, dx \,, \qquad \int e^{-2x} \sin(3x) \, dx \,.$$

Solution (i) From the formula sheet we obtain

$$\frac{d}{dx}\left(\cosh^{-1} x\right) = \frac{1}{\sqrt{x^2-1}} \,.$$

So for $I = \int \cosh^{-1} x \, dx$ we integrate by parts, with $u = \cosh^{-1} x \Rightarrow du = \frac{1}{\sqrt{x^2-1}} dx$ and $dv = dx \Rightarrow v = x$ so that

$$I \;=\; x\cosh^{-1} x - \int \frac{x}{\sqrt{x^2-1}} \, dx \;=\; x\cosh^{-1} x - \sqrt{x^2-1} + C \,.$$

Mostly done well.

(ii) Integrate by parts again: In

$$I = \int_0^1 3x^2 \tan^{-1} x \, dx$$

let $u = \tan^{-1} x \Rightarrow du = \frac{1}{1+x^2} dx$ and $dv = 3x^2 dx \Rightarrow v = x^3$, so that

$$I = x^3 \tan^{-1} x - \int \frac{x^3}{1+x^2} \, dx = x^3 \tan^{-1} x - \int x - \frac{x}{x^2+1} \, dx$$

using polynomial division, and finally

$$I \;=\; \left[x^3 \tan^{-1} x - \frac{x^2}{2} + \frac{1}{2}\ln(x^2+1)\right]_0^1 \;=\; \tan^{-1} 1 - \frac{1}{2} + \frac{1}{2}\ln 2 = \frac{\pi}{4} - \frac{1}{2} + \frac{1}{2}\ln 2 \,.$$

Many people got to $\int \frac{x^3}{1+x^2} \, dx$ and got stuck there.

(iii) To avoid integrating by parts two more times, we write

$$I = Im\left(\int e^{-2x}(\cos 3x + i \sin 3x) \, dx\right)$$

so integrate

$$I \;=\; \int e^{-2x} e^{i3x} \, dx \;=\; \int e^{(-2+3i)x} \, dx \;=\; \frac{e^{(-2+3i)x}}{(-2+3i)} + K$$

where K may be complex. To take the imaginary part, we write

$$\frac{e^{-2x}(\cos 3x + i \sin 3x)}{(-2+3i)} \frac{(-2-3i)}{(-2-3i)} \;=\; \text{real part} + i\frac{1}{13}\left(-3e^{-2x}\cos 3x - 2e^{-2x}\sin 3x\right)$$

so the imaginary part is $-\frac{e^{-2x}}{13}(3\cos 3x + 2\sin 3x) + C$.

Almost noone used this shortcut, opting instead to integrate by parts, twice. A lot of arithmetic errors.

(b) If $f(x) = \ln(1+x)$, show that its Maclaurin series is

$$f(x) \;=\; \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}\, x^n \;=\; x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \,,$$

and find the interval of convergence.

Solution Begin with $f(x) = \ln(1+x) \Rightarrow f(0) = 0$

$$f'(x) = \frac{1}{1+x} \Rightarrow f'(0) = 1$$
$$f''(x) = -\frac{1}{(1+x)^2} \Rightarrow f''(0) = -1$$
$$f'''(x) = \frac{2}{(1+x)^3} \Rightarrow f'''(0) = 2$$
$$f^{(4)}(x) = \frac{6}{(1+x)^3}\,.$$

We can see the pattern and spot that

$$f^{(n)}(x) = (-1)^{n+1}\frac{(n-1)!}{(1+x)^{n+1}} \Rightarrow f^{(n)}(0) = (-1)^{n+1}(n-1)!$$

and the series is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n \;=\; \sum_{n=1}^{\infty} (-1)^{n+1}\frac{(n-1)!}{n!}x^n \;=\; \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}\, x^n$$

as required. To find the interval of convergence we apply the ratio test:

$$\lim_{n\to\infty} \left| \frac{\frac{(-1)^{n+2}}{n+1}\, x^{n+1}}{\frac{(-1)^{n+1}}{n}\, x^n} \right| \;=\; \lim_{n\to\infty} \frac{n}{n+1}|x| \;=\; |x| < 1$$

for convergence, giving the interval of convergence $-1 < x < 1$. Testing at the endpoints we have

$$x = 1: \quad \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$$

which is the alternating harmonic series, and converges, and

$$x = -1 \quad \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(-1)^n}{n} = -\sum_{n=1}^{\infty} \frac{1}{n}$$

which is the harmonic series and diverges.

Hence the interval of convergence is $-1 < x \leq 1$.

A few people just wrote down the result from the formula sheet: no marks. Many used the ratio test correctly to find the interval of convergence, but then didn't evaluate, or evaluated incorrectly at $x = \pm 1$.

(c) Use the first three terms of the series obtained in (b), above, to estimate

$$\int_0^1 \frac{\ln(1+x)}{x}\,dx\,.$$

Use the remainder term to obtain a bound on the size of the error.

Solution: We take the first three terms and get

$$\int_0^1 \frac{\ln(1+x)}{x}\,dx \;\approx\; \int_0^1 \frac{1}{x}\left(x-\frac{x^2}{2}+\frac{x^3}{3}\right)\,dx \;=\; \left[x-\frac{x^2}{4}+\frac{x^3}{9}\right]_0^1 \;=\; 1-\frac{1}{4}+\frac{1}{9} \;=\; \frac{31}{36}\,.$$

We use the remainder term to obtain an error bound:

$$\int_0^1 \frac{1}{x}R_3\,dx \;=\; -\int_0^1 \frac{1}{x}\left(\frac{3!}{4!(1+c)^4}x^4\right)\,dx \;=\; -\int_0^1 \frac{1}{4(1+c)^4}x^3\,dx$$

As $0<c<x\le 1$, the integrand is maximal when $c=0$, so we obtain

$$|\text{Error}| \;<\; \frac{1}{4}\int_0^1 x^3\,dx \;=\frac{1}{16}\,.$$

Much confusion here. Many people did not read the question correctly, and did not get to $\int_0^1 \frac{1}{x}\left(x-\frac{x^2}{2}+\frac{x^3}{3}\right)\,dx$, *or tried* $\int_0^1 \left(x-\frac{x^2}{2}+\frac{x^3}{3}\right)\,dx$ *instead. A few tried to integrate* $\ln(1+x)/x$. *Of those that set it up correctly, many got to 31/36, but then evaluated the remainder term incorrectly and got another error bound.*

3. (a) Find the solution of the first-order ODE

$$\frac{dx}{dt} \;=\; \frac{-3t-x}{2x-2t}+\frac{3x}{2t}\;,$$

giving $x(t)$ as an explicit function of t.

Solution: Rewrite as

$$\frac{dx}{dt} \;=\; \frac{-3-\frac{x}{t}}{2\frac{x}{t}-2}+\frac{3}{2}\left(\frac{x}{t}\right)\;,$$

which is homogenous. Use the substitution $v=\frac{x}{t} \Rightarrow \dfrac{dx}{dt}=t\dfrac{dv}{dt}+v$ and obtain the separable ODE

$$t\frac{dv}{dt}+v=\frac{-3-v}{2v-2}+\frac{3}{2}v \;\Rightarrow\; t\frac{dv}{dt}=\frac{v^2-2v-3}{2v-2} \;\Rightarrow\; \int\frac{2v-2}{v^2-2v-3}\,dv \;=\; \int\frac{1}{t}\,dt$$

and integrate to get

$$\ln(v^2-2v-3)=\ln t \;+\; C \;\Rightarrow\; v^2-2v-3=Kt \quad (K=e^C\in\mathbb{R}) \qquad\qquad (*)$$

To obtain an explicit solution, complete the square:

$$(v-1)^2=4+Kt \;\Rightarrow\; \frac{x}{t}=v=1\pm\sqrt{4+Kt} \;\Rightarrow x=t\pm t\sqrt{4+Kt}$$

is the solution.

Mostly done correctly up to $(*)$, but many people then substituted $v=x/t$ into the quadratic and failed to get x as an explicit function of t.

(b) Show that the substitution $x=e^t$ transforms the Euler equation, a second-order linear ODE, with non-constant coefficients

$$x^2\frac{d^2y}{dx^2}+ax\frac{dy}{dx}+by=f(x)$$

into the second-order linear ODE with constant coefficients

$$\frac{d^2y}{dt^2}+(a-1)\frac{dy}{dt}+by=f(e^t)$$

[6]

Solution: Given

$$x=e^t \Rightarrow \frac{dx}{dt} \;=\; \frac{d^2x}{dt^2}=e^t=x\;,$$

we get

$$\frac{dy}{dx}=\frac{\frac{dy}{dt}}{\frac{dx}{dt}} \;\Rightarrow\; \boxed{x\frac{dy}{dx} \;=\; \frac{dy}{dt}}$$

and

$$\frac{d^2y}{dt^2} \;=\; \frac{d}{dt}\left(\frac{dy}{dt}\right) \;=\; \frac{d}{dt}\left(\frac{dy}{dx}\frac{dx}{dt}\right) \;=\; \frac{d^2x}{dt^2}\frac{dy}{dx}+\frac{dx}{dt}\frac{d}{dt}\left(\frac{dy}{dx}\right) \;=\; x\frac{dy}{dx}+x\frac{d}{dx}\left(\frac{dy}{dx}\right)\frac{dx}{dt} \;=\; x\frac{dy}{dx}+x^2\frac{d^2y}{dx^2}$$

giving $\boxed{x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dt^2} - \frac{dy}{dt}}$. Substituting the two boxed results into the ODE, we obtain

$$\frac{d^2y}{dt^2} - \frac{dy}{dt} + a\frac{dy}{dt} + by = f(e^t) \,,$$

as required.

Many good answers, but a lot of people got to the first boxed equation and then ran into trouble on the way to the second boxed equation, often just writing down the result.

(c) Given $u(x,y) = x\cosh y$ and $v(x,y) = x\sinh y$, if $f(u,v)$ is any function of u and v, use the chain rule to show that

$$\frac{\partial f}{\partial u} \;=\; \cosh y\,\frac{\partial f}{\partial x} \;-\; \frac{\sinh y}{x}\,\frac{\partial f}{\partial y}\,,$$

and find a similar expression for $\dfrac{\partial f}{\partial v}$. [7]

Solution: Given $u(x,y) = x\cosh y$ and $v(x,y) = x\sinh y$, we can write

$$u^2 - v^2 \;=\; x^2\cosh^2 y - x^2\sinh^2 y \;=\; x^2 \Rightarrow x \;=\; \sqrt{u^2 - v^2}\,,$$

and

$$\frac{v}{u} \;=\; \frac{x\sinh y}{x\cosh y} = \tanh y \;\Rightarrow y \;=\; \tanh^{-1}\left(\frac{v}{u}\right)\,.$$

and use the chain rule to obtain

$$\frac{\partial x}{\partial u} \;=\; \frac{u}{\sqrt{u^2 - v^2}} \;=\; \cosh y\,, \quad \text{and similarly} \quad \frac{\partial x}{\partial v} \;=\; -\sinh y$$

Likewise,

$$\frac{\partial y}{\partial u} \;=\; \frac{1}{1 - \left(\frac{v}{u}\right)^2}\left(-\frac{v}{u^2}\right) \;=\; -\frac{v}{u^2 - v^2} \;=\; -\frac{\sinh y}{x} \quad \text{and similarly} \quad \frac{\partial y}{\partial v} \;=\; \frac{\cosh y}{x}$$

Finally, use the chain rule to obtain

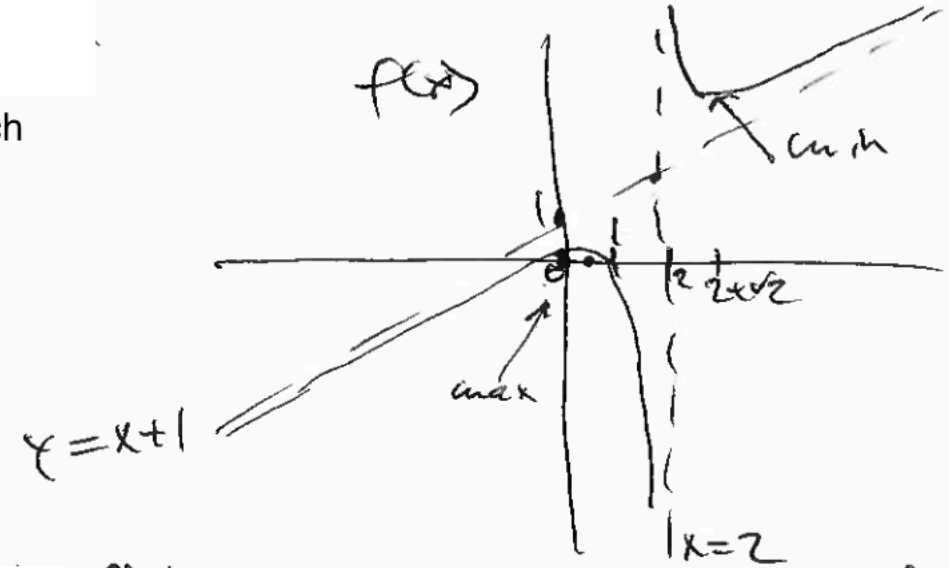
$$\frac{\partial f}{\partial u} \;=\; \frac{\partial f}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial u} \;=\; \cosh y\,\frac{\partial f}{\partial x} \;-\; \frac{\sinh y}{x}\,\frac{\partial f}{\partial y}\,,$$

as required. Similarly,

$$\frac{\partial f}{\partial v} \;=\; \frac{\partial f}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial v} \;=\; -\sinh y\,\frac{\partial f}{\partial x} \;+\; \frac{\cosh y}{x}\,\frac{\partial f}{\partial y}\,.$$

Few people rewrote to get $x(u,v)$ and $y(u,v)$. They instead applied the chain rule directly to u and v, differentiating w.r.t x, y. This meant solving simultaneous equations involving derivatives and functions: not easy. Of those that tried this, quite a few got the expression given for f_u, and then couldn't find the second for f_v. A lot of people tried to force an incorrect derivation into giving the right result.

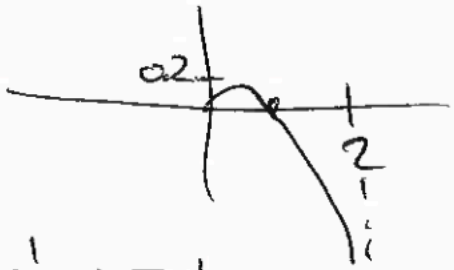
Q1b sketch



intercepts at $x=0$, $x=1$

$g(x)$ = width of $f(x)$ on $[0, 2]$ $\Rightarrow$

Q1c sketch

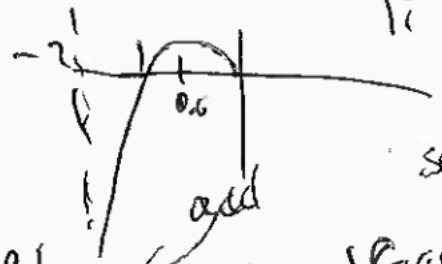


Value at max?

$$f(2-\sqrt{2}) = \frac{(2-\sqrt{2})(1-\sqrt{2})}{-\sqrt{2}}$$

$$= \frac{4-3\sqrt{2}}{-\sqrt{2}} = 3-2\sqrt{2} \approx 0.2$$

sketch $g(-x)$



add

$\rightarrow g_e = \frac{1}{2}(g(x)+g(-x))$



subtract

$g_o = \frac{1}{2}[g(x)-g(-x)]$ $\Rightarrow$

